

Computer Networks(CS3001)

Marking Scheme with Rubrics

Q1:

[10 x 0.5 = 5 Marks]

Accuracy (0.25 Marks per Answer): Whether the answer is true or false.

Justification (0.25 Marks per correct Answer):

Give your answer as true or false to show your agreement or disapproval to the following along with one sentence justification. Note: No points will be awarded without justification.

a) Multiplexing and de-multiplexing occur only at the transport layer.

Answer: False – Multiplexing and de-multiplexing happen at multiple layers of the networking stack.

b) In TCP, flow control ensures that the receiver does not overwhelm the sender.

Answer: False – TCP's flow control mechanism prevents the sender from sending more data than the receiver can handle.

c) The UDP header includes a checksum field for error detection and correction.

Answer: False – The UDP header has a checksum field that helps detect errors in transmission.

d) The maximum TCP segment size can be of 64KB only.

Answer: True – The maximum receive window field is of 16-bit only.

e) TCP uses a cumulative acknowledgment strategy.

Answer: True – TCP acknowledges the highest byte received in order and implicitly acknowledges all previous bytes.

f) The DNS client retransmits the query only after a timeout if no response is received.

Answer: True – DNS uses UDP, which doesn't have any acknowledgment mechanism.

g) TCP protocol ensures that packets are never lost in the network.

Answer: False – TCP protocol minimizes packet loss but cannot guarantee that packets will never be lost.

h) The TCP timeout value must be fixed for reliable communication.

Answer: False – TCP dynamically adjusts the timeout value based on estimated round-trip time (RTT).

i) The transport layer is responsible for routing packets between networks.

Answer: False – Routing is handled by the network layer, not the transport layer.

j) The sliding window protocol improves efficiency by allowing multiple packets to be in transit before receiving an acknowledgment.

Answer: True – Sliding window protocols increase efficiency by sending multiple packets before requiring an acknowledgment.

Q2:

[2.5 + 2.5 = 5 Marks]

Illustrate using the timing sequence diagram, the following:

a) TCP 3-way Handshake

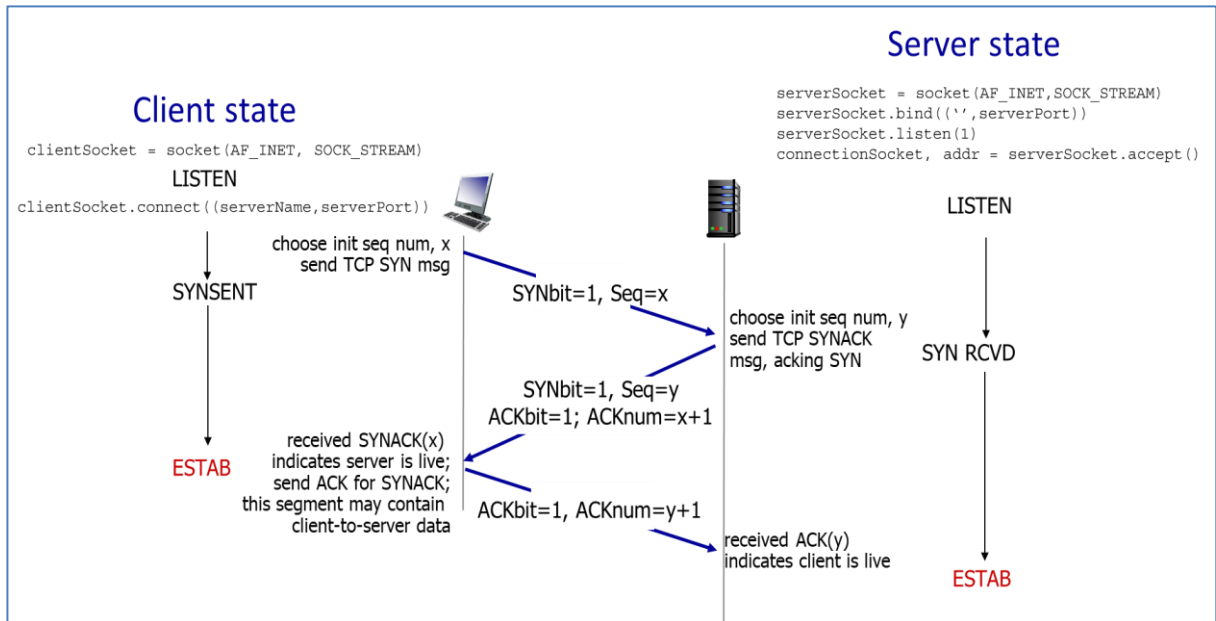
SYNbit = 1, Seq = x, 0.5 marks

SYNbit = 1, Seq = y 0.5 marks

ACKbit = 1, ACKnum = x+1 0.5 marks

ACKbit = 1, ACKnum = y+1 0.5 marks

LISTEN > SYNSENT / SYN RCVD > ESTAB 0.5 marks

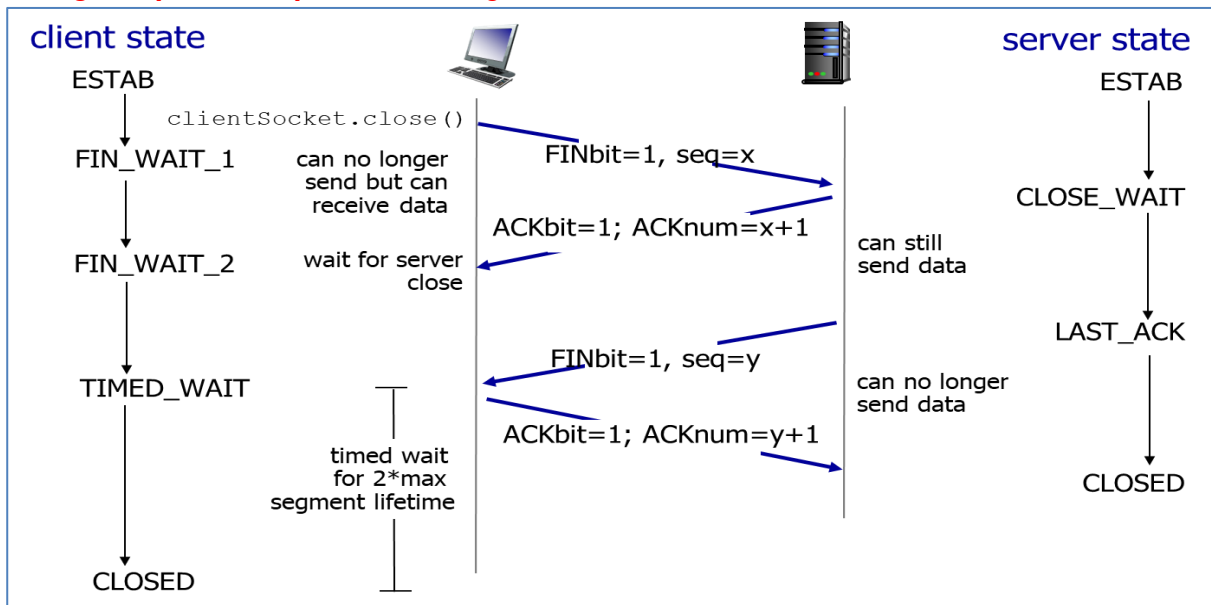


b) Closing a TCP Connection

FIN Message Details (0.5 Marks): Proper sequencing and acknowledgement of the FIN messages.

State Transitions (1.5 Marks): Correct state transitions for both client and server.

Correct Diagram (0.5 Marks): Accurate diagram of the TCP connection closure.



Q3:

[2.5 + 2.5 = 5 Marks]

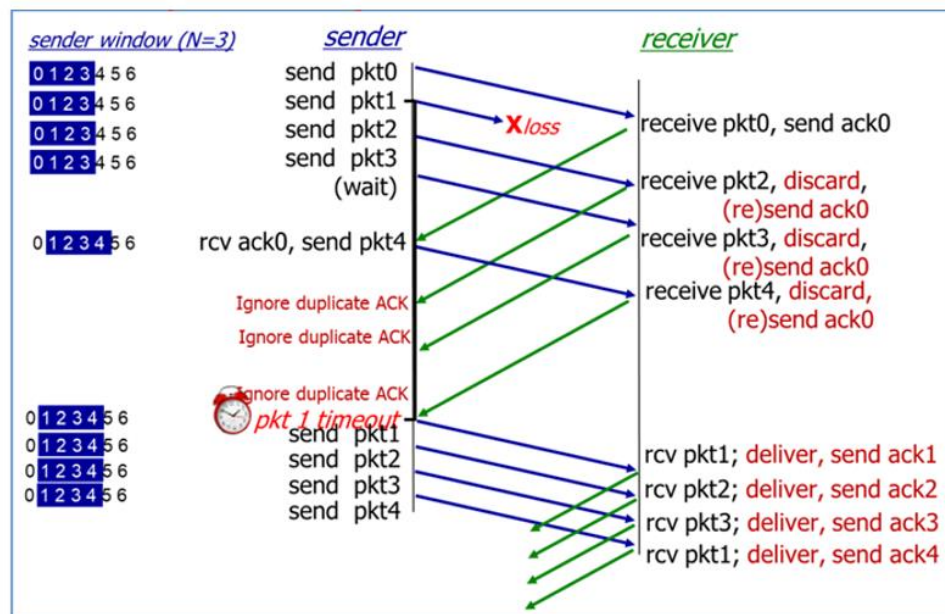
- a) A sender transmits packets using Go-Back-N (GBN) with a window size of 4 and a timeout period of 10 ms. The sequence numbers available are 0, 1, 2, 3, 4, 5, 6. The sender transmits packets 0, 1, 2, and 3. However, packet 1 is lost during transmission. The receiver uses cumulative acknowledgment (ACK). Draw a time sequence diagram and show the GBN operation for the given scenario.

Assume: Each packet takes 5 ms to be transmitted and acknowledged. The sender starts transmission at $T = 0$ ms. The timeout occurs at $T = 10$ ms.

Correct Window Sequence Numbers (0.5 Marks): Proper sequencing of window numbers.

Handling of Out-of-Order Packets (0.5 Marks): Accurate representation of packet discarding.
Handling of Duplicate Acknowledgments (0.5 Marks): Correct treatment of duplicate acknowledgments.
Resending of Lost Packets (0.5 Marks): Correct representation of resending the lost packet.
Successful Delivery of Resent Packets (0.5 Marks): Correct acknowledgment and successful delivery.

Solution:



Time (ms)	Sender Actions	Receiver Actions	ACKs
0	Send pkt 0	Receive pkt 0	Send ACK 0
5	Send pkt 1	** (Lost) **	
10	Send pkt 2	Receive pkt 2	** (Out of order - Discard) **
15	Send pkt 3	Receive pkt 3	** (Out of order - Discard) **
20	** Timeout for pkt 1 **		
	Retransmit pkt 1	Receive pkt 1	Send ACK 1
25	Retransmit pkt 2	Receive pkt 2	Send ACK 2
30	Retransmit pkt 3	Receive pkt 3	Send ACK 3

- The receiver discards out-of-order packets because GBN does not buffer them.
- The sender retransmits all packets from the lost packet onwards.
- Cumulative ACKs are used, meaning the receiver only acknowledges up to the last correct packet in order.

b) A sender transmits packets using Selective Repeat (SR) with a window size of 4 and a timeout period of 10 ms. The available sequence numbers are 0, 1, 2, 3, 4, 5, 6. The sender transmits packets 0, 1, 2, and 3. The receiver uses individual acknowledgment (ACK). Draw a time sequence diagram and show the SR operation for the given scenario.

Assume: Each packet takes 5 ms to be transmitted and acknowledged. The sender starts transmission at $T = 0$ ms. The timeout occurs at $T = 10$ ms.

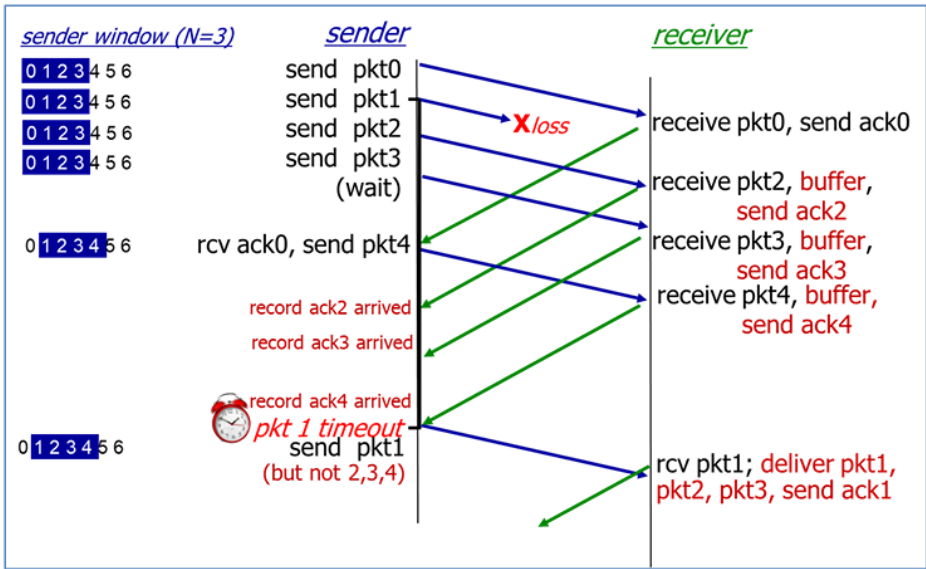
Correct Window Sequence Numbers (0.5 Marks): Proper sequencing for SR protocol.

Buffering of Out-of-Order Packets (0.5 Marks): Accurate representation of buffering.

Recording of Out-of-Order Acknowledgments (0.5 Marks): Correct handling of out-of-order acknowledgments.

Resending Only Lost Packets (0.5 Marks): Correct depiction of only the lost packet being resent.

Successful Delivery (0.5 Marks): Successful delivery after retransmissions.
 Solution:



Time (ms)	Sender Actions	Receiver Actions	ACKs
0	Send pkt 0	Receive pkt 0	Send ACK 0
5	Send pkt 1	**(Lost)**	
10	Send pkt 2	Receive pkt 2	**(Buffered)**
15	Send pkt 3	Receive pkt 3	**(Buffered)**
20	***Timeout for pkt 1**		
	Retransmit pkt 1	Receive pkt 1	Send ACK 1
25		Deliver pkt 1, pkt 2, pkt 3	Send ACK 2, 3

- The receiver buffers out-of-order packets (pkt 2 and pkt 3).
- The sender retransmits only the lost packet (pkt 1) instead of all packets.
- Once pkt 1 is received, the receiver delivers buffered packets and sends ACKs.